CHAPTER 5 · CONCEPT 2 · WORKSHEET

Practice — Contact Forces: Muscular Force and Friction

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ **[1]**

A force that acts only when objects are in contact is a —

(a) non-contact force (b) contact force (c) magnetic force (d) gravitational force

Q2 · MCQ **[1]**

Friction always acts —

(a) in the direction of motion (b) opposite to the direction of motion (c) upwards (d) at random

Q3 · MCQ **[1]**

Friction is greater on a —

(a) smooth surface (b) rough surface (c) wet glass (d) polished floor

Q4 · MCQ **[1]**

Chewing food and the heart pumping blood are examples of —

(a) magnetic force (b) muscular force (c) electrostatic force (d) friction

Q5 · ASSERTION–REASON **[1]**

A: Friction is a contact force.
R: It arises due to irregularities of two surfaces in contact.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK **[1]**

Friction arises due to the _________ present on the surfaces in contact.

____

Q7 · FILL IN THE BLANK **[1]**

Aeroplanes and ships are given a _________ shape to reduce friction from air/water.

____

Short answer (2 marks each)

Q8 **[2]**

What is a contact force? Give two examples.

Q9 **[2]**

What is friction and in which direction does it act?

Q10 **[2]**

On what does the force of friction depend? How does roughness affect it?

Long / application (3 marks each)

Q11 **[3]**

A box is pushed and slides across glass, wood and sand, stopping at different distances each time. Explain these observations using friction.

Q12 **[3]**

Explain why friction is called a contact force and give one situation where it is helpful and one where it is a nuisance.

Check yourself → Answer key